DEPARTMENT OF ELECTRICAL
AND COMPUTER ENGINEERING

RICARDO CAMACHO QUADRIO DA CUNHA MONTEIRO

Master in Electrotechnical and Computer Engineering

BASKETBALL MOTIONS RECOGNITION AND INJURY PREVENTION

APPLICATION OF A NON DISRUPTIVE SMART WEARABLE IN BASKETBALL

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING

NOVA University Lisbon

Draft: *July 31, 2025*

DEPARTMENT OF ELECTRICAL
AND COMPUTER ENGINEERING

BASKETBALL MOTIONS RECOGNITION AND INJURY PREVENTION

APPLICATION OF A NON DISRUPTIVE SMART WEARABLE IN BASKETBALL

RICARDO CAMACHO QUADRIO DA CUNHA MONTEIRO

Master in Electrotechnical and Computer Engineering

Adviser: Anikó
Full Professor, NOVA University Lisbon

Co-adviser: Rodolfo Oliveira
Full Professor, NOVA University Lisbon

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING

NOVA University Lisbon
Draft: *July 31, 2025*

Contents

List of Figures

1 Introduction

The world is always looking for newer and better ways to improve efficiency and durability of everything including ourselves. Wearable technology is an example of this research and its popularity is increasing over time. These devices have a great wide-range of functions that can help in medical diagnosis, injury prevention, workout detection, gait measurement, sports performance, sleep analysis, among others.

Wearables have experienced a great growth in the last few years in every possible way. These devices, paired with smartphones, have revolutionized how we as a society live our day to day lives and how we improve sports. It's no secret that statistics play a big part in today's sports landscape and this it is only getting more complex with the progression of time.

Basketball is one of the most popular sports worldwide and it is constantly evolving and so does the related technology alongside it. Gathering data from the players is crucial to improve the efficiency of their practices and their game. As of today, coaches rely in video analysis to help their players with tactical movements and technical skills but this method has its limitations such as blocked views or difficult angles for the recording device especially in a five vs five scenario. This is a strong suit of wearable sensors since they can capture data continuously and with the right design, be unnoticeable for the user and other players.

1.1 Wearables

A Wearable is any smart device with sensors that can be worn, embedded in clothing, accessories or in the body or even adhered to or tattooed on the skin. There are many different types of Wearables such as Smart Socks, Smart Insoles, Smartwatches, Smart Rings amongst others. These devices are able to measure many different types of data such as position, pressure, brightness, velocity, heart rate and many others [19].

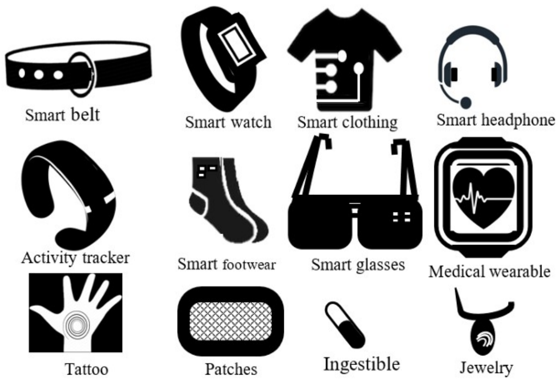

Figure 1.1: Different wearables [8].

Nowadays there are many different wearables that can collect the same type of data. With this in mind, the choice for the right device doesn't only revolve around what type of information it can collect. Portability, processing power, accessibility, battery life are some other important characteristics.

1.1.1 Applications

Wearables provide data useful for many applications. Their portability, affordability, easy connection to a mobile phone and non-prescription nature of these devices makes them an easy choice for several uses. Amongst others the main ones are Health, Activity Recognition and Sports and Tracking/Localization [8].

1.1.1.1 Health

In the health sector wearables have gained a lot of importance over the last few years. With smart devices, doctors and patients can get a continuous health monitoring which lead to better diagnostics and disease and injury prevention. The data provided by the devices is also used to design insoles, shoes and other objects that aid health care and injury prevention [15]. Besides this, wearables are also being used by people to track their own health without any prescription.

Construction companies have also been investing in wearables to prevent injuries in their staff when working with heavy loads. In 2020 there were 619 million people affected by lower back pain globally and the number is estimated to increase to 843 million by 2050 [17].

1.1.1.2 Tracking and Localization

Knowing the position of a person or an animal is important in many occasions. From tracking animals to understand their behaviour to simply finding where a lost elder is, this category has many useful applications.

This devices tend to have a great battery life efficiency since they have to last long periods of time without being charged.

1.1.1.3 Activity Recognition and Sports

Even tho this section has a lot of overlapping technology with the health segment, its use cases go beyond the health category. Besides the day-to-day monitoring these devices can also recognize and monitor your physical activities.

With information about timings, amount of force, position and others, coaches and players can prevent injuries and improve their skills.

1.1.1.4 Others

Wearables are also used for education, gaming, virtual and augmented reality, law enforcement besides other applications.

1.1.2 Wearables in Basketball

Basketball was created in 1891 by a Canadian physical education professor named James Naismith in Springfield, Massachusetts as an indoor game to avoid the rain.

Sensors initially became part of the game through the ball manufacturing process in the 1950s to improve safety and standardization of its pressure. In the 1980s the players started using blood oxygen and hear rate monitors in their training to improve strength and conditioning. The development of image sensors led to the recording and live broadcasting of basketball games in the 1990s and the 2000s. Displacement and acceleration sensors starting being used in the 2010s to monitor the players' shooting form and other movements. As mentioned in section 1.1 wearable technology is rapidly evolving and it is the next big step for sports statistics and performance improvement. [25].

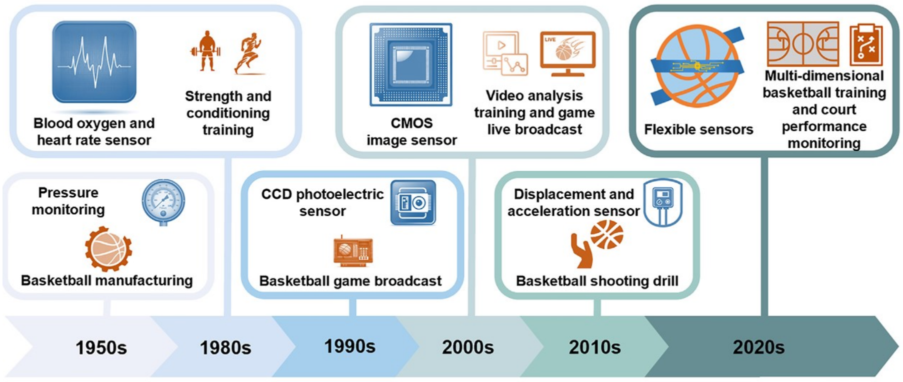

Figure 1.2: [25, Timeline of sensing technology intertwined with basketball sport].

Dave Brailsford popularized the concept "Marginal gains" which consists in improving every single detail, no matter how small in order to get get better. The biggest obstacle in this philosophy is quantifying and identifying what we can improve. The answer to this distress are Wearables [14].

For this purpose the technology used consists mostly of micro-electromechanical systems such as inertial measurement units (IMU's), networked sensor suites and exoskeletons. Risk assessment shares some of its data with performance enhancement devices but it also works with pressure sensors and IMU's that host several sensors (e.g., accelerometers, gyroscopes, and magnetometers) [13].

1.2 Project Introduction and Proposition

When comparing team sports to single-activity recognition, due to the involvement of two parties trying to better the other one and its unpredictability, team sports have a greater complexity. They also involve a lot of high intensity short periods of time and stress, induced by some game situations specially at game climax [12].

For basketball in particular, the movements consist in a lot of short sprints, quick changes of pace and direction, lateral movement and high jumps (vertical and horizontal) and its high tense moments are more common at the end of games (fourth quarter), specially at the last minutes.

Basketball and more specifically, the NBA are becoming more popular globally with each passing year. This leads to more data and more efficient ways to capture it. The most common method of doing it is through video recording but these devices are expensive and only the teams with high budgets can afford them [20]. The NBA collects statistics through several high definition 3D cameras than can record the whole field. This technology is called SportVU and has been introduced to the league in September of 2013 [18].

For this project two devices consisting of an Arduino and an Emg Sensor will be used to collect data. This is an initial design that can be improved by implementing these sensors into an insole or a sock making it as comfortable and as unnoticeable as possible. An android application will be developed as an interface for the user and the database used.

Even tho another other devices have increased precision and other types of data, in basketball, the only allowed wearables are the ones that can be hidden and won't cause any disturbances or interferences in the game such as injuring someone.

2

State Of The Art

2.1 Movement Recognition Wearables

As mentioned before in the introduction, there are many different wearables capable of collecting the same type of data but in different ways. For gait monitoring there are several different types of sensors used in wearables that can be separated in five different categories [4]:

1. Capacitive Sensors - Consisting in two conductive plates separated by an insulating layer. When force is applied, they develop a voltage variation.

2. Resistive Sensors - These are the most common sensors. Their resistance varies with the applied pressure.

3. Optoelectronic Sensors - Composed by a transmitter and a receiver (a laser or a LED and a photodiode). When a force is applied, the light that reaches the photodiode varies proportionally.

4. Piezoresistive and Piezoelectric Sensors - These sensors use the piezoelectric effect. They convert the changes in applied pressure to electrical current.

2.2 IMU - Inertial Measurement Unit

An IMU is a device consisting of a combination of accelerometers and gyroscopes and sometimes magnetometers tracking motion.

2.3 Plantar Pressure Wearables

Plantar pressure wearables are innovative devices designed to monitor and analyze the distribution of pressure across the soles of the feet. By embedding sensors into insoles, smart socks, or even shoes, these wearables capture detailed data on foot loading patterns, which is critical for understanding gait, balance, and overall foot health. Plantar pressure data offers valuable insights for applications in sports performance, injury prevention, rehabilitation, and the management of conditions such as diabetes, where pressure distribution can indicate areas at risk for ulcers or other complications. They are also useful for better footwear designs or orthosis [1, 6, 21, 22].

As of today there are four different kinds of devices that can measure plantar pressure [4]:

1. Platform Systems - Known as the most reliable method for measuring plantar pressure, platform systems offer high accuracy and detailed spatial resolution. Their main limitation is a lack of portability, restricting their use to controlled environments like labs and hospitals.

2. In-Shoe Systems: These systems provide greater mobility and flexibility, with optimized circuit design, power efficiency, and communication technology, and they're more affordable than platform systems. They record plantar pressure inside the shoe and allow for extensive data collection across multiple steps, enabling long-term gait analysis indoors and outdoors. However, they are less accurate than platform-based measurements.

3. Smart Wireless Insoles: Unlike platform and in-shoe systems, smart wireless insoles are cordless, eliminating the need for wires to connect sensors or data acquisition units. While in-shoe systems aren't ideal for extended outdoor use, smart insoles can be used both indoors and outdoors. They often include Bluetooth or WiFi for data transmission and a power source. However, the added elastic layer they create inside the shoe can be several millimetres thick, potentially distorting the accuracy of foot pressure data. They are also relatively expensive and not suited for daily wear.

4. Smart Socks: These fabric-based systems integrate sensors directly into the textile, which avoids the challenges of the previous systems. However, they require handmade or complex manufacturing techniques, which is a drawback.

In basketball, smart insoles and smart socks are particularly well-suited wearables due to their portability, flexibility, and adaptability to both indoor and outdoor environments. Unlike larger, stationary measurement systems, these wearable devices are attached to the athlete, providing continuous and real-time data regardless of the setting.

Smart insoles and socks, when embedded with pressure sensors, can track a player's foot pressure distribution, stability, and impact forces with each step, jump, or landing. This information is crucial for basketball players, as it helps to analyze their movement patterns, improve balance, reduce injury risk, and enhance overall performance. The insole's thin profile and ability to fit inside regular shoes also ensure they don't interfere with the player's comfort or mobility. Similarly, smart socks, which incorporate sensors directly into the fabric, offer flexibility and comfort while capturing dynamic foot pressure and movement data. Their lightweight design and close contact with the foot provide detailed insights into the nuances of foot pressure changes during the quick movements in basketball. Both smart insoles and socks also benefit from wireless connectivity, allowing data to be transmitted to a smartphone or tablet, enabling real-time monitoring and analysis. This flexibility and data accessibility are particularly valuable in basketball training and performance analysis, as players and coaches can assess and adjust techniques in real time, whether in an indoor or outdoor court.

2.3.1 Restraints

Wearables have to be light and comfortable since they should be unnoticeable and not disturb the user's activities. This means that these gadgets have low processing capacities and their sensors monitor the data at lower frequencies then non wearable devices. Due to their size they often use miniature sensors that can be sensitive to placement and alignment. Inaccurate placement, external interference, or device movement can cause errors, reducing the reliability of collected data. Additionally, in applications where precise measurements are essential, the smaller sensors in wearables can't match the accuracy of larger, lab-based equipment.

Wearables are often exposed to a wide range of environmental conditions, including sweat, water, dust, and physical impacts. Ensuring these devices remain durable and function properly under such conditions is a challenge, as components like sensors and circuits need to be robust enough to withstand daily use and environmental exposure.

Due to their lack of memory and processing power, the data collected from the devices has to be sent to another device for storage or further processing. The main communication protocols with low energy consumption are Bluetooth Low Energy (BLE) and WiFi for close data transmission. These methods can be impacted by limited range, interference, and connectivity issues, especially in crowded or signal-dense environments, which can disrupt real-time data streaming.

Since these devices are quite small, one of their main restraints is the battery life. With every action taken by the device, whether it's collecting data with a sensor, processing that data or sending it somewhere else, energy is always spent. Depending on their purpose, wearables may have to last up until one year without being charged [24]. To overcome this challenge some devices rely on energy harvesting components to charge the device such as solar power, motion energy, etc [6].

Minding the others problems already mentioned, wearables have low security so they are relatively easy targets due to their poor encryption and protection [8]. Due to the lack of privacy of the athlete's data and since wearable technology can possibly obstruct and injury someone, wearables still are'nt accepted in many sports (in federated play), basketball included. Even tho they are't allowed in games, there aren't any rules for practices and there are some teams that are taking advantage of these devices [16, 7].

2.3.2 Communication Protocols

Wearable technology relies heavily on efficient communication protocols to ensure seamless data exchange between devices, applications, and networks. These protocols define the rules for data transmission, enabling wearables to monitor health, track activity, and connect with other devices in real time. Given the constraints of wearables, communication protocols play a critical role in balancing performance with power efficiency. For this project two kinds of protocols will be discussed: Bluetooth Low Energy and WiFi.

2.3.2.1 Bluetooth Low Energy (BLE)

Bluetooth Low Energy (BLE) is a short-range communication technology known for its low power consumption. Classified as a Wireless Personal Area Network (WPAN), BLE is widely used in various wearable applications, from diabetes monitoring devices to smartwatches. The original insole design employed BLE for communication between the insole and the paired mobile application. BLE's primary design objectives are to enable low-power communication, particularly for peripheral devices. It is optimized for affordability and simplicity, aligning well with many developer requirements and contributing to its swift adoption. The main difference between BLE and the classic Bluetooth is their transmission rates. BLE transmits at around 2000 Kbps while Bluetooth Classic transmits at around 2-3 Mbps making its energy consumption around two times bigger than BLE. This is the reason BLE is the usual choice for IOT devices. Besides this difference BLE works the same way as Bluetooth Classic [11, 9].

2.3.2.2 WiFI

WiFi is a widely used wireless Local Area Network (LAN) technology, primarily aimed at connecting mobile devices to the internet. Its broad adoption means it's available in most homes, businesses, and many public spaces. WiFi uses variations of the IEEE 802.11 protocol depending on the generation, with the latest generation, WiFi 6 (802.11ax), approved in September 2024. Operating on the 2.4 GHz or 5 GHz frequency bands, WiFi 6 also supports wider channels, optimized mainly for video streaming. WiFi delivers high data speeds and easy setup for access points, making it a mature and well-proven technology. However, its range is somewhat limited indoors due to high signal absorption, though range improves in open, line-of-sight environments. Due to its relatively high power consumption, WiFi is not ideal for direct communication in IoT devices [11, 9].

2.3.3 Gait Cycle

The Gait Cycle is the sequence of movements that one leg takes in each step in walking or running. It is the whole moving process from the first point of contact between a foot and the ground until it reaches the ground again (see figure2.1) [11, 23].

The usual method for measuring Gait Cycle is with a platform and a multi-camera setup since it is the most precise and reliable. However, this method is very expensive and can only be applicable indoors in a secure environment. An insole or a smart sock aren't as reliable but they compensate by being cheaper, more comfortable for the user and can be worn in their activities.

Although the Gait Cycle varies from person to person (minding age, body type and others) and even in each step each person takes, there are many tendencies for normal gait. With this in mind these patterns can be used to predict and understand what forces are applied in a gait cycle [11].

There are three foot's directions(see figure2.2): Vertical which is the direction perpendicular to the foot's palm. Fore-aft that follows the line between the toe to the heel. And the lateral direction which goes from one side of the foot to the other [11].

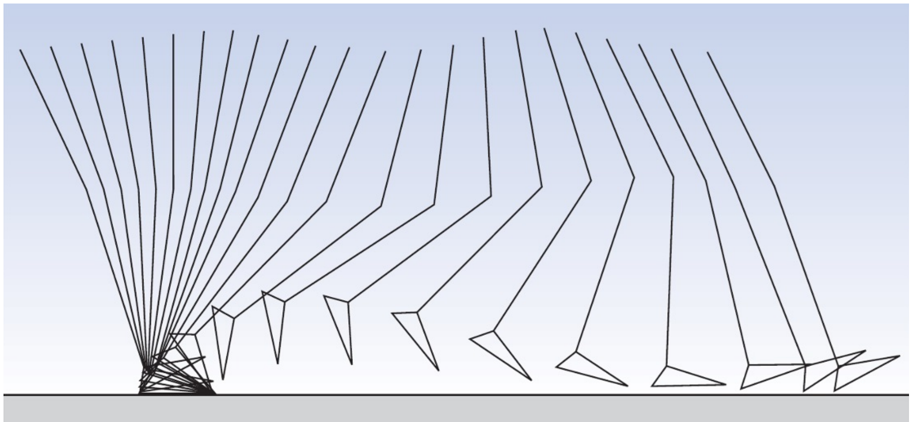

Figure 2.1: The gait cycle of a leg with with a cycle time of 0.92s. 40 ms interval between lines [10].

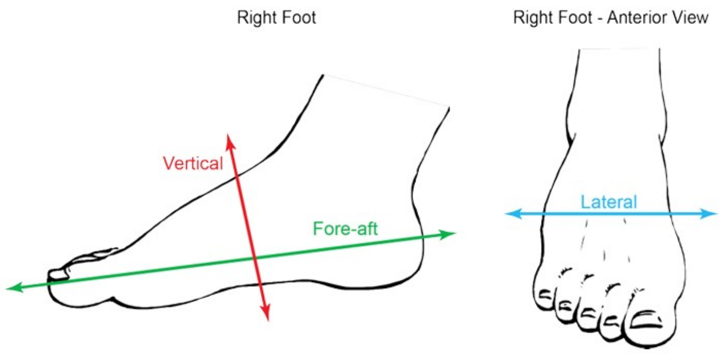

Figure 2.2: The three foot's directions [10].

There are two different stages for this motion in slower gaits and an extra stage in faster gaits [2]:

- Stance Phase - when the foot is in contact with the ground. This represents about 60% of the normal gait cycle.
- Swing Phase - when the foot is in the air. It normally represents the remaining 40% of the gait cycle.
- Flight Phase - this phase only occurs in faster gaits such as running. It refers to the moments where neither foot is in contact with the ground.

2.3.4 Other Movements

Basketball involves many other distinct movements besides gait. These motions are very complex making them difficult to accurately identify but they are essential for injury prevention and for the performance enhancement analysis. In this project data will be collected from three groups of movements with increasing complexity.

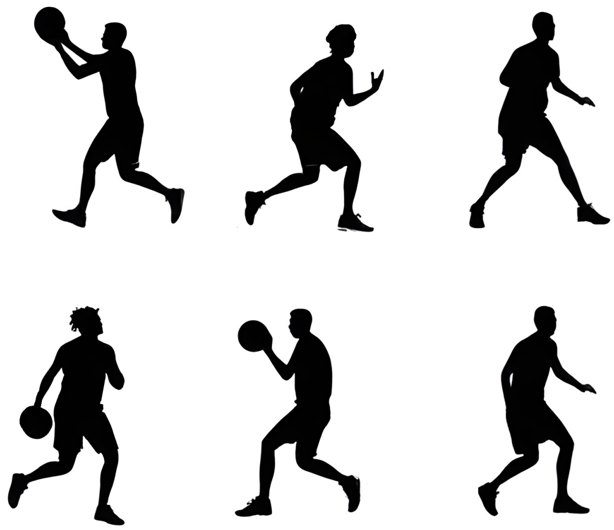

Figure 2.3: Some basketball stances.

2.3.4.1 Basic Movements

- Layup - A layup is normally a two step shot and it requires a step-by-step increase in pressure, usually peaking in the final step before jumping.
- Crossover Dribble - In a crossover dribble, the player shifts their weight rapidly from one foot to the other, using pressure changes to decelerate in one direction and quickly accelerate in the opposite direction.
- Sprint(fast gait) - Basketball players frequently sprint down the court, which involves a pattern of quick, alternating footfalls with consistent gait cycles on each foot.
- Defence Slide/Stance - The defensive slide requires the player to move laterally while maintaining a low stance.
- Boxout - The act of securing a positing usually near the basket to deny the rebound for the opponent.
- Rebound Jump - Rebounding requires explosive jumping to reach the ball, sometimes with minimal preparatory movement, other times with a box out.

- Cutting - Cutting involves a sharp change in direction with a great speed shift. It is an offball movement.

2.3.4.2 Intermediate Movements

- Pivot - A pivot involves planting one foot firmly while the other foot moves to change direction.
- Change of Direction in Defence - A player changes his direction while sliding.
- Screen - Similar to the boxout but it is done anywhere on the court to force a defensive player to change his path.
- Jumpshot - The act of shooting the ball. The jump shot is a fundamental shooting technique in basketball, characterized by a quick dip in pressure as the player lowers into a squat before takeoff, followed by a burst of force as they push off the ground to jump.

2.3.4.3 Advanced Movements

- Eurostep - This movement is a specific type of layup where the player takes the first step in a direction and the second step to another direction.
- Step Back - The step back is a Jump Shot whe the player takes a step back before shooting the ball.
- Failed Landing (Incomplete layup) - Any type of fall characterized by the sudden fall of pressure and following.

2.4 Related Work

There are many studies on the topic of measuring plantar pressure through an insole. Most of them mostly study the gait cycle while running or walking with similar objectives. Baitong Wang et al. [23] developed an insole with 8 pressure sensors and motion track sensors and with this system managed to clearly identify gait with only the pressure sensors.

Elizabeth S. Chumanov et al. [3] tracked the global position of the pressure on the soles of the feet throughout steps in walking and running. This data can be used for improvement in foot-floor contact models used to simulate human locomotion and in least squares inverse dynamics.

For sports like basketball, the available motion recognition research was done through video recording or smart wearable devices such has wristbands. There is very few studies on the use of insoles for this purpose.

Thomas Holleczek et al. [5] design a pair of smart socks with three pressure sensors each for the use case of snowboard. Even tho it isn't a sport as dynamic and unpredictable as basketball, it still is very relevant for this project. They worked on activity recognition focusing on recognizing if the user was riding backside or riding frontside (with his back to the bottom of the hill or the top respectively). It is pointed that the recognizability of three basic standing postures was explored and that they were distinguished with perfect accuracy mentioning that it can be interesting for gait analysis.

Qing F. Zhou et al. [20] utilized insoles with a motion sensor with the purpose of getting a high accuracy for recognizing three basketball motions: Dribbling, jumping and turning around. For this data processing the K-means algorithm was utilized. Two years later they continued their work and managed to identify five new different footwork movements. In this version a multiple sensor system was proposed and three different classification methods were discussed [18]: K-Nearest neighbours (KNN), Support vector machine (SVM) and Convolutional Neural Networks (CNN).

Catarina Amaro et al. [1] studied the differences in five different movements in basketball between two different moments in a season. The study presented no significant changes in mean peak pressure between the two different times but it did specify that experienced players had lower plantar pressure values in their movements and lower symmetry index. This means that there is room for improvement based on the values read for the less experienced players. It was also mentioned that wear of the shoes may have influenced the slight changes observed in the study. Each player wore their own shoes and this was pointed as a variable for the plantar pressure measured.

2.5 Relevant data for analysis

Wearable devices capture large volumes of raw data, often gathering detailed metrics like pressure distribution, movement patterns, and force dynamics. However, to extract meaningful insights, this raw data must first be transformed into a more organized, manageable format suited for analysis. Preprocessing includes filtering out noise, aligning data with relevant time frames, and structuring it into clear categories or features, which simplifies identifying trends and patterns. By converting this data into a cleaner, standardized format, we can more effectively apply analysis techniques, uncovering insights into biomechanics, performance, and even long-term health outcomes.

2.5.1 Average Pressure

Foot therapists currently rely on average pressure to determine if a support sole is effective and where additional relief might be needed. This average provides an interpretable view of pressure distribution during the gait cycle. However, averaging all values across all frames would dilute important information because of the many zero values present throughout the cycle. To avoid this, only non-zero values are averaged for each sensor. Yet, this approach introduces another challenge: a cell with few non-zero readings could appear similar to one with many readings. To improve clarity, cells are excluded from analysis if they don't exceed a minimum threshold of non-zero measurements for that sensor[11].

2.5.2 Peak Pressure (PP), Pressure Time Integral (PTI) and Mean Peak Pressure (mPP)

When assessing plantar pressure, the primary parameters of interest are peak pressure (the highest recorded plantar pressure), pressure time integral (typically defined as the area beneath the peak pressure-time curve), and mean peak pressure (mPP). However, studies have shown that mPP and PTI are closely related, making it unnecessary to measure both.

When there are differences in these measurements between the feet, it can mean that there is an inadequate distribution of forces and it can lead to injury. So a symmetry index defined by the next equation that compares the plantar pressure on both feet will be calculated.

$$SI = \frac{2 \cdot |R - L|}{(R + L)} \times 100$$

If SI is zero then the feet are doing equal work, if it's under 10% it is acceptable, otherwise it means the movement is causing more distress in one leg than the other leading to a loss of bone mass density in the respective leg and joints [1].

2.6 Data Analysis

By leveraging advanced data analysis techniques, such as machine learning algorithms and statistical modelling, pressure insole data can reveal subtle patterns and correlations that inform training adjustments, injury risk, and performance improvements. There are many different algorithms with different objectives. In this section some of them will be explained:

2.6.1 K-nearest-neighbours classification

The K-nearest-neighbours is one of the simplest supervised machine learning algorithms. It's a classification algorithm which uses proximity to assigning each point to the class with the most similar elements within the K-nearest-neighbours [9, 2].

As an example observe figure2.4. If someone would have to assign a class (green diamonds, blue squares or red circles) to the cross in the figure, they would probably say it belongs to the green diamonds since they are the closest objects. But if instead someone had to assign a class for the star, it would be a much harder task with a naked eye. In this case two factors make that decision: Which are the closest objects (depending on the weight k) and how many of each class are there in that group [9]. In other words imagine a small circle with its center on the object that we want to classify and if it isn't touching other know objects imagine it growing until it does.

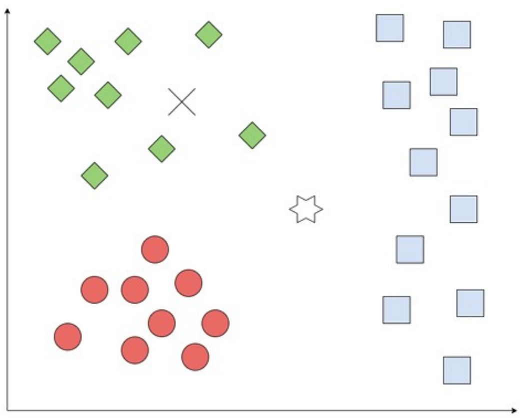

Figure 2.4: K-nearest-neighbours example [9]

2.6.2 Dynamic Time Warping

The dynamic Time Warping (DTW) algorithm transforms unsynchronized time series or time series with different lengths allowing a dynamic comparison between them. This algorithm calculates the best possible alignment between two different series dependent on time providing a measurement of how well aligned (or not) they are [9].

2.6.3 Regression analysis

Regression analysis' purpose is to find a continuous outcome variable, usually y, through one or more predictor variables, normally x (if only one). The simplest of these models is the linear regression model that relates a linear relationship between x and y. This equation can be expressed as [9]:

$$y = \beta + \alpha \cdot x + \epsilon$$

In this equation y is the output variable, β is the starting point (x=0), α is the regression coefficient associated with the predictor variable x and ϵ is the residual error. An extended version of this equation with n predictor variables is:

$$y = \beta + \alpha_1 \cdot x_1 + \alpha_2 \cdot x_2 + \alpha_n \cdot x_n + \epsilon$$

2.6.4 Recursive Feature Elimination

Recursive Feature Elimination (RFE) selects features by progressively narrowing down the set of available features. Each feature is assigned a weight, and those with the lowest absolute weights are removed from the set. In each iteration, a single feature is removed, and this process continues on the reduced set until the target number of features is reached [2].

2.6.5 Principal Component Analysis

Principal Component Analysis (PCA) reduces the dimensionality of data by applying a linear transformation based on singular value decomposition. This transformation projects the data into a lower-dimensional space, creating a new set of variables that are ordered by their importance [2].

2.6.6 Linear Discriminant Analysis

The Linear Discriminant Analysis (LDA) model fits a Gaussian distribution to each class, assuming a shared covariance matrix across all classes. This model reduces input dimensionality by projecting data onto the directions that best separate the classes. Its goal is to enhance separation between classes by maximizing the scatter between them while minimizing scatter within each class [2].

3 Work Proposal

This project begins with the prototype design and the development of the interface app. Then data of the different movements is collected and the ML model is developed. The last goal is to implement the best model on to the app giving the use real-time feedback and analysis of his movements.

3.1 Project Description

For this project, for each foot, an Arduino Nano 33 BLE will be used. These devices have an integrated IMU each that captures acceleration and orientation and they work with BLE Besides the IMU, an Emg sensor will collect data from the muscles to provide better recognition and a better insight for injuries.

3.1.1 Prototype

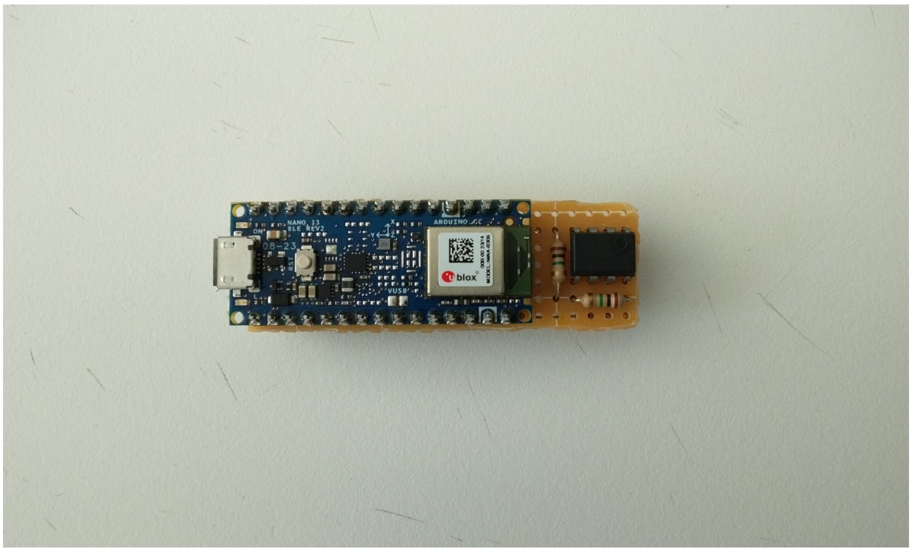

Figure 3.1: Topside of the Prototype - Arduino

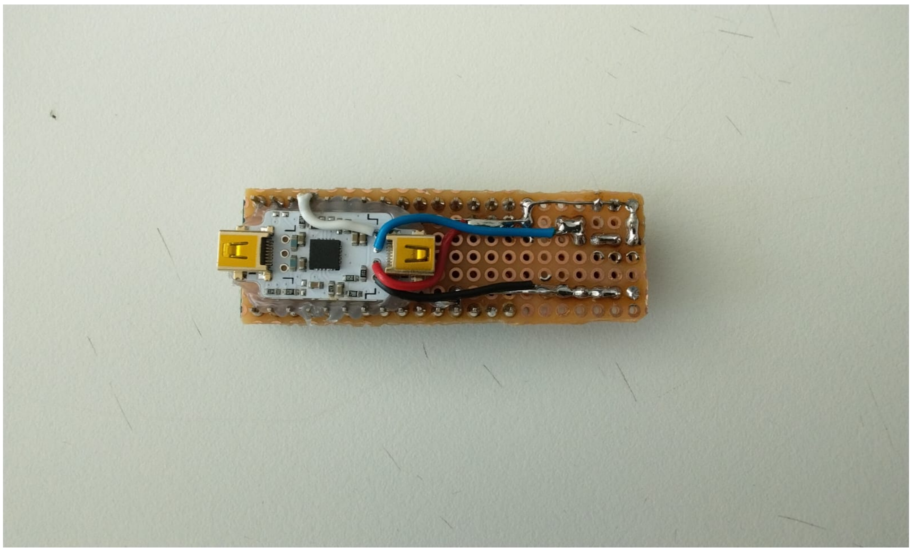

Figure 3.2: Bottom of the Prototype - Emg

4

Conclusion

With the ongoing development of wearable technology, particularly smart insoles and socks, there is also a significant advancement in basketball motion recognition and injury prevention. Even tho wearables have several restraints, this technology is the most appropriate for basketball motion recognition and injury prevention and its biggest step back is the prohibition in official matches.

By capturing real-time data on pressure distribution, foot orientation, and movement dynamics, these wearables allow for a detailed analysis of key basketball actions like jump shots, layups, pivots, and lateral movements, providing valuable insights into each player's unique biomechanics for the players and coaches.

Bibliography

[1] C. Amaro et al. "Plantar Pressure Evaluation during the Season in Five Basketball Movements". In: *Applied Sciences* 10 (2020-12), p. 8691. DOI: 10.3390/app10238691 (cit. on pp. 7, 14, 15).

[2] B. F. D. Cardoso. "Sensors fusion and movement analysis for sports performance optimization". MA thesis. Faculdade De Engenharia Da Universidade Do Porto, 2017. URL: https://repositorio-aberto.up.pt/bitstream/10216/108411/2/226423.pdf (cit. on pp. 11, 16, 17).

[3] E. Chumanov, C. Remy, and D. Thelen. "Tracking the Position of Insole Pressure Sensors during Walking and Running". In: () (cit. on p. 14).

[4] A. Drăgulinescu et al. "Smart Socks and In-Shoe Systems: State-of-the-Art for Two Popular Technologies for Foot Motion Analysis, Sports, and Medical Applications". In: *Sensors* 20.15 (2020). ISSN: 1424-8220. DOI: 10.3390/s20154316. URL: https://www.mdpi.com/1424-8220/20/15/4316 (cit. on p. 7).

[5] T. Holleczek et al. "Textile pressure sensors for sports applications". In: *SENSORS, 2010 IEEE*. 2010, pp. 732–737. DOI: 10.1109/ICSENS.2010.5690041 (cit. on p. 14).

[6] M. Hubl et al. "Embedding of wearable electronics into smart sensor insole". In: *2016 IEEE 18th Electronics Packaging Technology Conference (EPTC)*. 2016, pp. 597–601. DOI: 10.1109/EPTC.2016.7861550 (cit. on pp. 7, 9).

[7] *Injury Prevention Technology? Hawks Players Are Wearing It*. 2015. URL: https://www.nba.com/hawks/features/injury-prevention-technology-hawks-players-are-wearing-it (cit. on p. 9).

[8] F. John Dian, R. Vahidnia, and A. Rahmati. "Wearables and the Internet of Things (IoT), Applications, Opportunities, and Challenges: A Survey". In: *IEEE Access* 8 (2020), pp. 69200–69211. DOI: 10.1109/ACCESS.2020.2986329 (cit. on pp. 2, 9).

[9] C. R. H. Kjær and C. B. Henriksen. "Smart Sole- Classi cation of lifts using IoT technology". MA thesis. Technical University of Denmark, 2021 (cit. on pp. 10, 16, 17).

[10] D. Levine, J. Richards, and M. Whittle. *Whittle's gait analysis*. 2012-07. ISBN: 9780702042652 (cit. on p. 11).

[11] C. Lundberg. "Data collection, management and visualisation for IoT healthcare devices". MA thesis. Technical University of Denmark, 2020 (cit. on pp. 10, 15).

[12] J. Lutz et al. "Wearables for Integrative Performance and Tactic Analyses: Opportunities, Challenges, and Future Directions". In: *International Journal of Environmental Research and Public Health* 17.1 (2020). ISSN: 1660-4601. DOI: 10.3390/ijerph17010059. URL: https://www.mdpi.com/1660-4601/17/1/59 (cit. on p. 4).

[13] S. McDevitt et al. "Wearables for Biomechanical Performance Optimization and Risk Assessment in Industrial and Sports Applications". In: *Bioengineering* 9.1 (2022). ISSN: 2306-5354. DOI: 10.3390/bioengineering9010033. URL: https://www.mdpi.com/2306-5354/9/1/33 (cit. on p. 4).

[14] G. M. Migliaccio, J. Padulo, and L. Russo. "The Impact of Wearable Technologies on Marginal Gains in Sports Performance: An Integrative Overview on Advances in Sports, Exercise, and Health". In: *Applied Sciences* 14.15 (2024). ISSN: 2076-3417. DOI: 10.3390/app14156649. URL: https://www.mdpi.com/2076-3417/14/15/6649 (cit. on p. 4).

[15] H. Nagano and R. K. Begg. "Shoe-Insole Technology for Injury Prevention in Walking". In: *Sensors* 18.5 (2018). ISSN: 1424-8220. DOI: 10.3390/s18051468. URL: https://www.mdpi.com/1424-8220/18/5/1468 (cit. on p. 2).

[16] *NBA, NBPA Use Wearables To Study Player Health and Wellness*. 2024. URL: https://athletechnews.com/nba-nbpa-use-wearables-to-study-player-health-wellness/ (cit. on p. 9).

[17] W. H. Organization. *Low back pain*. 2023. URL: https://www.who.int/news-room/fact-sheets/detail/low-back-pain (cit. on p. 3).

[18] M. Peng, Z. Zhang, and Q. Zhou. "Basketball Footwork Recognition using Smart Insoles Integrated with Multiple Sensors". In: *2020 IEEE/CIC International Conference on Communications in China (ICCC)*. 2020, pp. 1202–1207. DOI: 10.1109/ICCC49849.2020.9238862 (cit. on pp. 5, 14).

[19] A. Ç. Seçkin, B. Ateş, and M. Seçkin. "Review on Wearable Technology in Sports: Concepts, Challenges and Opportunities". In: *Applied Sciences* 13.18 (2023). ISSN: 2076-3417. DOI: 10.3390/app131810399. URL: https://www.mdpi.com/2076-3417/13/18/10399 (cit. on p. 2).

[20] S. Shi et al. "Utilize Smart Insole to Recognize Basketball Motions". In: *2018 IEEE 4th International Conference on Computer and Communications (ICCC)*. 2018, pp. 1430–1434. DOI: 10.1109/CompComm.2018.8780718 (cit. on pp. 5, 14).

[21] A. M. Tan et al. "Development of a Smart Insole for Medical and Sports Purposes". In: *Procedia Engineering* 112 (2015). 'The Impact of Technology on Sport VI' 7th Asia-Pacific Congress on Sports Technology, APCST2015, pp. 152–156. ISSN: 1877-7058. DOI: https://doi.org/10.1016/j.proeng.2015.07.191. URL: https://www.sciencedirect.com/science/article/pii/S187770581501440X (cit. on p. 7).

[22] Y. Tian et al. "Deep-learning enabled smart insole system aiming for multifunctional foot-healthcare applications". In: *Exploration* 4 (2023-11). DOI: 10.1002/EXP.20230109 (cit. on p. 7).

[23] B. Wang et al. "FreeWalker: a smart insole for longitudinal gait analysis". In: *2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2015, pp. 3723–3726. DOI: 10.1109/EMBC.2015.7319202 (cit. on pp. 10, 14).

[24] J. Williamson et al. "Data sensing and analysis: Challenges for wearables". In: *The 20th Asia and South Pacific Design Automation Conference*. 2015, pp. 136–141. DOI: 10.1109/ASPDAC.2015.7058994 (cit. on p. 9).

[25] H. Zeng et al. "Intelligent health and sport: An interplay between flexible sensors and basketball". In: *iScience* 27.3 (2024), p. 109089. ISSN: 2589-0042. DOI: https://doi.org/10.1016/j.isci.2024.109089. URL: https://www.sciencedirect.com/science/article/pii/S2589004224003109 (cit. on p. 4).